Semantified CEUR-WS proceedings publishing in Wikidata knowledge graph

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Abstract. CEUR Workshop Proceedings (CEUR-WS) has been a non-profit publisher for scientific workshop and conference proceedings since 1995. For more than 30 years it has used HTML and PDF as the single source of truth for the metadata required by indexers such as DBLP and the German national library DNB. Scholars have long craved persistent identifiers for the core entities involved: event series, events, proceedings, papers, editors, authors, institutions, and publishers, with the goal of allowing semantic handling and querying with modern tools.
In this work we show how we enabled the semantification using Wikidata as the target knowledge graph infrastructure. The 2015 Semantic Publishing Challenge queries are finally permanently available with up-to-date metadata. The Scholia portal provides aspect-oriented and user-friendly access and is widely used.
We further point out the challenges of aligning the digital, social, and physical aspects of the ongoing transformation and outline a roadmap to use the available queries to bootstrap a new RDF-based metadata-first publishing approach that will mine metadata from proceedings matter and generate the HTML from it instead of the other way around.

Keywords: Knowledge Graph · Linked Data · Metadata Extraction · Publishing · Semantic Web · Wikidata

1 Introduction

Fig. 1. Most relevant entities for scientific publishing (abstract schema). Source: CR research wiki [10].

Traditional publishing workflows often handle the metadata of the needed core entities (as shown in Figure 1) in an unFAIR way, giving low priority to

Fig. 2. SWAT4HCLS2024 example graph walk

the Findable, Accessible, Interoperable, and Reusable (FAIR) 3.1, Single Point of Truth (SPoT), and Single Source of Truth (SSoT) principles. Therefore the opportunities to collect the metadata during the lifecyle of an event [15] are missed. As technical editors of CEUR-WS we have been feeling this pain for over a decade. This work shows how the HTML/PDF SSoT of CEUR-WS at the ceur-ws.org domain (SPoT) has been used to create a Knowledge Graph representation of the relevant entities in Wikidata between 2022 and 2026 as a step in this direction, thus allowing portals such as Scholia to present the RDF query results in an aspect-oriented and user-friendly way.

1.1 CEUR Workshop Proceedings

Since 1995 CEUR Workshop Proceedings (CEUR-WS) (https://ceur-ws.org/) has published more than 4,200 Volumes and over 90,000 papers as a free online service for publishing proceedings of scientific workshops (and smaller conferences) in computer science, operated by a team of unpaid volunteers.

Technically, all data and metadata of the proceedings are directly represented as a static website using HTML, PDF and HTTP and FTP protocols as provided by the workshop editors along the CEUR-WS "how to submit" guide [5].

The metadata is available after some delay of weeks or months through indexing services such as DBLP [21] and K10plus [30][4] and at the archive of the TIB `https://www.tib.eu/`.

After multiple unfinished attempts to make the metadata of the CEUR-WS platform available for semantic analysis and querying we report on the successful use of Wikidata and Scholia as the knowledge graph platform to provide FAIR metadata.

Doing so was harder than one would expect - the devil is in the details.

1.2 Contributions of this Paper

The core contributions presented in this work are:

1. To provide the tools, infrastructure and approaches to consistently supply high quality FAIR metadata (Semantification) for CEUR-WS as detailed in section 3.
2. Splitting the metadata for the three main entities: Event, Event Series and Proceedings as a major step towards improving the metadata quality and FAIRness.
3. Providing a bootstrapping [3,8] approach to shift the single point of truth from the current HTML/PDF files to a RDF ready ontology based format as outlined in 5.

[4] `https://dblp.org/`, `https://opac.k10plus.de`

These contributions are valuable for other publishing use cases that struggle with the same pain points.

2 From Semantic Publishing Challenge 2014 via Text2KG 2023 to Production 2026

The Semantic Publishing Challenge [20,7,28] started in 2014 with Task 1 calling for "answering queries related to the quality of workshops by computing metrics from data extracted from their proceedings, also considering information about persons and events".

The relevance of the original set of 20 queries for Task 1 was subjectively rated by us, resulting in a list sorting the queries by priority[5]. The most relevant queries and the 5 queries Q1.5, Q1.12, Q1.13, Q1.16 and Q1.17 that rely on the main index have been taken as the definition of done. We reported on the successful Wikidata export of all CEUR-WS proceeding and event metadata in 2023 at the TEXT2KG Workshop [15]. The SPARQL queries[6] are available as named parameterized queries. The snapquery project [14] was motivated by this set of queries and was a candidate middleware for the Scholia project [34].

3 CEUR-WS Semantification in Wikidata

Figure 3 gives an overview of the CEUR-WS Semantification workflow.

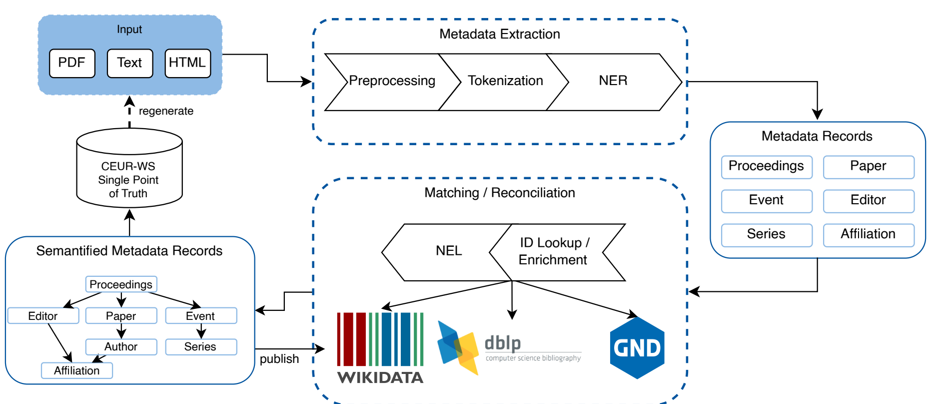

Fig. 3. Workflow of the CEUR-WS Semantification

5 https://cr.bitplan.com/index.php/List_of_Queries
6 https://cr.bitplan.com/index.php/Semantic_Publishing_Challenge_Queries

3.1 Making CEUR-WS FAIR

The FAIR principles [19,33] with Persistent Identifiers (PIDs) and rich metadata as core components are being adopted by an increasing number of publishing organizations.

To make CEUR-WS metadata FAIR a set of relevant queries should be supported, as derived from the original set of queries of the 2014 Semantic publishing challenge as outlined in Section 2. The metadata should be provided following an ontology that is based on established ontologies. The manual and automatic curation of entries should be possible with public access for all stakeholders, e.g., editors, authors, organizers, publishers and indexers. The infrastructure should be stable and there should be sufficient trust in its long-term availability An open source non-profit infrastructure is preferred since this is also the mode of operation of CEUR-WS.

Given the HTML/PDF/text input of CEUR-WS, the corresponding Knowledge Graph needs to be created and the single-point-of-truth computer readable metadata be separated from the different representations such as HTML so that the above requirements are fulfilled.

Both the HTML and PDF encoding of the original scientific content are structured for the purpose of optimizing the display / output on paper or screens; therefore, there is a structure loss compared to what was originally available in the text document processor files that the authors might have been using [9]. Most scholars are not aware of this loss in the daily use of published content just because the documents are optimized for display and human consumption [35,2]. For the metadata extraction and use in knowledge graphs the difference is sometimes disastrous, e.g., when a simple text from a PDF document can not be extracted any more due to exotic styling and formatting or simply because only a scanned graphic image version of an older document is available that needs optical character recognition to extract the textual content.

The metadata needed for creating the knowledge graph is only available in natural language/text form and follows rules that have been changed multiple times during the history of CEUR-WS. From 2013 to 2026, there have been 33 different versions[7] of the index file template with no proper tracing of what was changed from version to version. The index and volume HTML files were often edited manually, leading to minor, undocumented differences even within these versions. The usage of the different page versions follows a long-tail Zipfian distribution with 5 versions covering 60% of all volumes.

3.2 Extracting metadata from HTML and PDF targeting Wikidata

The pyCEURmake [12] tool was created to parse the main HTML index as well as the individual proceedings HTML pages vastly improving on the earlier simple scraper by Nielsen [23]. The results have been cached in JSON and SQLite storage for quick access. Location entries have been consolidated using

[7] https://cr.bitplan.com/index.php/CEUR-WS_Versions

the geograpy3 library [27]. Author, Editor and Paper details have been aligned with the DBLP SPARQL endpoint.

Cermine [29] and GROBID [22] have been run on the papers to provide additional metadata references.

For each core entity we analyzed the availability of Wikidata properties by using our wdgrid analysis tool [11] [17]. This tool also generates naive or sophisticated SPARQL queries. We aligned the queries with the stated needs of the Semantic Publishing Challenge.

snapquery [14] was used to make sure the queries are compatible with different endpoints and do not only run on the official blazegraph based Wikidata query endpoint.

ceur-spt [13] is the prototype for showing the benefits of semantification by improved linking and navigation capabilities compared to the current static CEUR-WS HTML page.

4 In-Use: Community Impact - WikiCite, Scholia, CEUR-WS

WikiCite is an initiative to create a bibliographic database based on Wikidata [31].

Scholia [25] is a portal serving wikidata content for the needs of scholars.

WikiCite, Wikidata, Scholia and the CEUR-WS stakeholder community are overlapping communities. There are over 100,000 distinct authors and over 6,500 distinct editors with digital traces in the DBLP CEUR-WS index entries.

4.1 Scholia

It is based on a set of over 380 named parameterized SPARQL queries organized around 40 different aspects. The Wikidata Scholarly Articles Subgraph Analysis [1] showed that roughly half of the 17 billion triples that form the main graph belong to scholarly articles and their linked entities. The Wikimedia foundation needed to perform a graph split due to a 4 Terabyte maximum storage limit of the Blazegraph backend in use. The Scholia community decided to use QLever [4] as the new backend and make all queries SPARQL1.1 compliant [34]. Note that before the graph split there was a hold on adding more scholarly articles to Wikidata which was the reason we did not mass-insert CEUR-WS paper metadata yet. Writes to Wikidata are performed through the officially approved bot account *User:CEUR-WS* [6], for which the community permission was obtained in 2023 [32].

4.2 WikiCite

WikiCite [31] provides the bibliographic backbone on which CEUR-WS metadata is contributed to Wikidata. Figure 6 shows the distribution of Wikidata editors who have created or revised CEUR-WS *proceedings* items (Q1143604

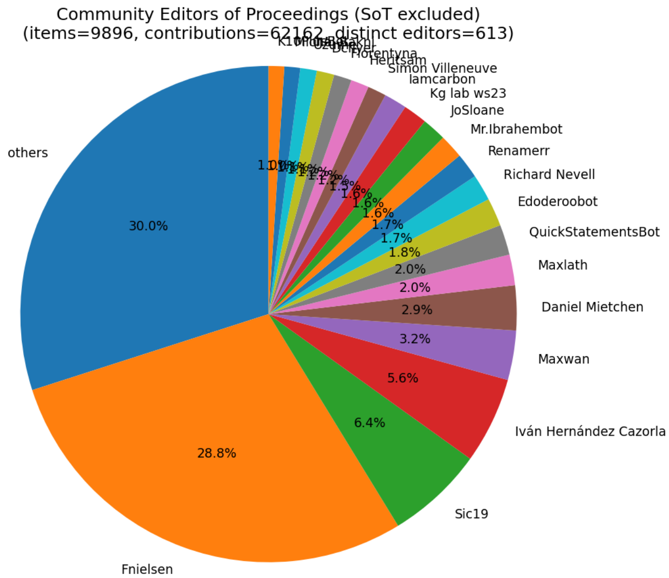

Fig. 4. Distribution of WikiCite community editors contributing to CEUR-WS proceedings items (Q1143604 (Q1143604)[10]) on Wikidata.

(Q1143604)[8]). The long-tail distribution illustrates that while the officially approved *User:CEUR-WS* bot [6] accounts for the bulk of systematic mass contributions, a broad community of individual WikiCite editors continuously curates, corrects and enriches the proceedings metadata. This overlap of bot-driven bulk ingestion and human community curation is a core strength of the WikiCite approach and a prerequisite for the Scholia panels discussed in Section 4.4 below. The plot was generated with `ceur-wd-contributions` [12] by traversing the revision history of all CEUR-WS proceedings items on Wikidata.

4.3 Scholia

Table 1 shows 12 SPARQL-driven panels on the Scholia event aspect [25]. All queries are enabled by this work and rendered identically by the legacy Blazegraph endpoint at `https://scholia.toolforge.org/event/Q124669329` and by the post graph-split QLever endpoint [4,34] at `https://qlever.scholia.wiki/event/Q124669329`, illustrated for SWAT4HCLS 2024 (Q124669329 (Q124669329)[11]). All 12 panels survived the 2025 Wikidata graph split unchanged; only the underlying engine and the SPARQL dialect (1.1 compliant) differ.

Figure 5 shows the user-friendly display of the coauthorship as graph visualization.

[8] https://www.wikidata.org/wiki/Q1143604
[11] https://www.wikidata.org/wiki/Q124669329

Table 1. Scholia SWAT4HCLS 2024 panels based on SPARQL queries

#	Panel/Query	style	entities
1	`data`	header	event
2	`people`	table	scholar
3	`co-authors`	graph	scholar
4	`sponsors`	table	institution
5	`timeline`	timeline	event
6	`topic-scores`	table	
7	`uses`	table	
8	`recent-publications`	table	paper, scholar
9	`proceedings`	table	proceedings, paper
10	`presentations`	table	
11	`related-events-timelocation`	table	event
12	`related-events-people`	table	event,scholar

4.4 WikiCite

WikiCite [31] provides the bibliographic backbone on which CEUR-WS metadata is contributed to Wikidata. Since the CEUR-WS Wikidata bot was activated the community members contributed to the bot created entries.

Figure 6 shows the distribution of Wikidata editors who have created or revised CEUR-WS *proceedings* items (Q1143604 (Q1143604)[12]). The long-tail distribution illustrates that while the officially approved *User:CEUR-WS* bot [6] accounts for the bulk of systematic mass contributions, a broad community of individual WikiCite editors continuously curates, corrects and enriches the proceedings metadata. This overlap of bot-driven bulk ingestion and human community curation is a core strength of the WikiCite approach and a prerequisite for the Scholia panels discussed in Section 4.4 below. The plot was generated with `ceur-wd-contributions` [12] by traversing the revision history of all CEUR-WS proceedings items on Wikidata.

5 Single Point of Truth Bootstrapping

6 Lessons Learned

6.1 Queries as first-class citizens

The Semantic Publishing Challenge and the Scholia portal both use queries as the core structuring elements. Using the guiding principle "If you can not query it it does not exist" changes the view on how to provide metadata. While traditionally schemas and ontologies are created first to allow querying for use-case specific areas of interest, the query-first and query as first-class citizens approach describes the information needs of stakeholders as queries. Having the

[12] https://www.wikidata.org/wiki/Q1143604

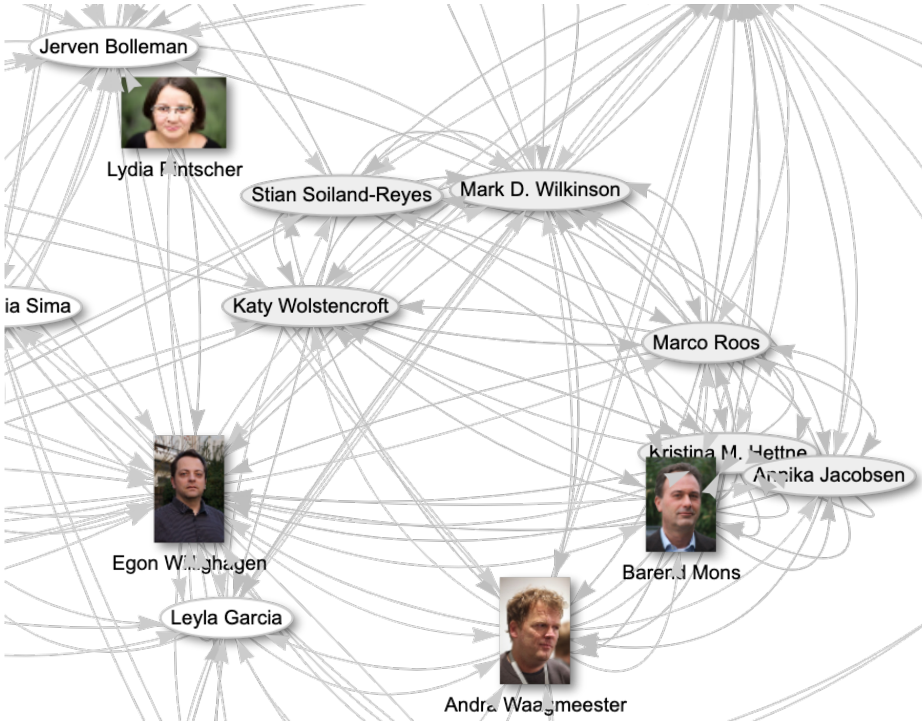

Fig. 5. SWATH4HCLS 2024 Coauthor graph visualization

queries as named parameterized technology independent entities themselves is helpful [14]. Nielsen's Synia [24] approach shows a lightweight application of this idea.

A main success factor for this work was being able to apply metadata to the queries. This way the queries could be ordered by task and/or relevance and linked to their respective implementation. These links were just motivations - as first class citizens the queries are available with proper URIs as persistent identifiers and part of a meta-knowledge graph that is useful for the further semantification steps.

6.2 Property creation as a community process

Wikidata properties such as P11633 (colocated with)[15] undergo a Wikidata's property proposal process[16] process before a new property is created. As described in [15] the creation of property proposal[17] took some effort and led to

[15] https://www.wikidata.org/wiki/Property:colocated with
[16] https://www.wikidata.org/wiki/Wikidata:Property_proposal
[17] https://www.wikidata.org/wiki/Wikidata:Property_proposal/colocated_with

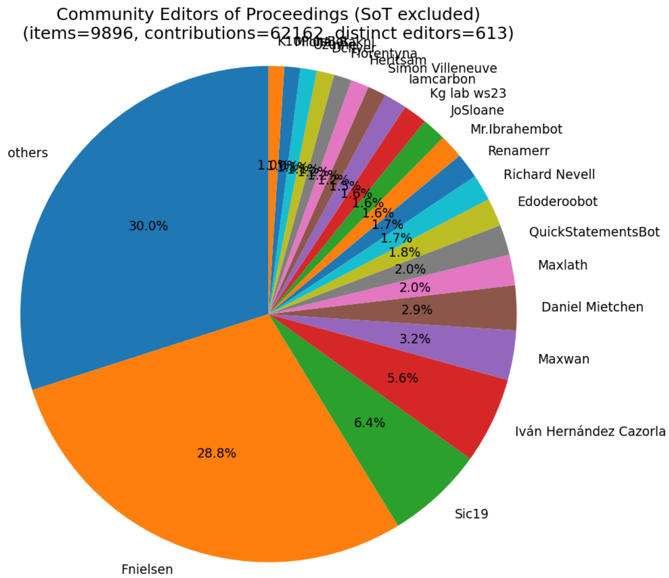

Fig. 6. Distribution of WikiCite community editors contributing to CEUR-WS proceedings items (Q1143604 (Q1143604)[14]) on Wikidata.

the clarification that the property is asymmetric. The property was considered as highly relevant and well defined.

6.3 Legal aspects of Publishing Personal Data of Scholars

The proposed workflow calls for publishing personal data of authors and editors early in the life cycle of an event. According to the legal advisors we asked, this is legal since a combination of conditions (Art. 6 GDPR) is met: Explicit consent is given by the stakeholders (Art. 6(1)(a)), the legitimate interest of the stakeholders is pursued (Art. 6(1)(f)), the data might be public already, the processing is necessary for scientific research (Art. 6(1)(e), Art. 89). The strongest condition is that there might be a legal obligation (Art. 6(1)(c)) as would be the case if publishing via major print publishers that have to provide a copy of the proceedings to the German National Library according to German law. Consent, Public Data, and Contractual Necessity are crucial for scholarly publishing.

6.4 Iterative introduction beats big bang

The challenge in introducing the new CEUR-WS workflow is to find a sweet spot between effort and benefit [26]. Given the long-tail distribution of issues of ever more corner and exotic cases that have shown up in the analysis of the existing data and workflow there is potentially a huge list of "small" problems that would require an inadequately high amount of effort to be solved systematically. The "sweet spot" is in solving only the most relevant problems and not fixing some of the exotic details at all or doing it manually. Therefore we will avoid a "big bang" introduction, which would replace the old approach in one single big (and therefore very risky) step. Instead we propose to do smaller changes gradually. As a first step, the main index.html shall be split by years (with a link to the old legacy complete list). Then, the per-year list may be generated from the single point of truth data. This will be continued iteratively, going from the current year back to a point where it seems no longer reasonable to spend further effort.

7 Roadmap: Metadata first Publishing

The coverage of the core entities Event Series, Papers, Scholars and Institutions needs to be improved based on consolidating and completing the RDF data of the downstream indexing services at DBLP and German National Library (DNB). The holy grail of this effort will be to change the CEUR-WS single point of truth from the current HTML/PDF base to a workflow in which the metadata is mined from the proceedings matter. The move of the CEUR-WS infrastructure from RWTH Aachen to Gesellschaft für Informatik in 2026 is an opportunity to make further steps in this direction. The CEUR-WS semantify project [18] is the open source effort pursuing this goal.

8 Conclusion

CEUR-WS proceedings metadata is systematically made available in Wikidata via the CEUR-WS bot. The relevant Task 1 Named Parameterized Queries of the Semantic Publishing Challenge of 2014/2015 are now available on different endpoints namely the SPARQL1.1 compatible QLever endpoint. The Scholia portal provides production access to queries based on and extending this set. The Wikidata, WikiCite, Scholia and CEUR-WS stakeholder communities have appreciated and contributed to this work.

8.1 Supplemental Material Statement

Source code for the prototypes and semantification tools is available via the repositories of the github ceurws organization[18]. Extended supplemental material — including the schema diagrams (Figures 1 and 2) in their editable Graphviz form, the SPARQL queries derived from the Semantic Publishing Challenge, and background research — is provided by the CR research wiki [10]. The LaTeX sources, bibliography, build pipeline and the `AGENTS.md` governance policy for AI-assisted editing of this paper are available at [16]. The cited software artefacts are released under Apache-2.0: `ceur-spt` [13], `pyCEURmake` [12], `snapquery` [14], and `wdgrid` [11].

[18] https://github.com/ceurws

8.2 Acknowledgements

We would like to thank Jakob Voß for helping with the k10plus matching and creating the Wikidata property colocated with (P11633)[19] in due time.

Thomas Hoeren and Jonas Kuiter of ITM, Münster kindly gave hints for legal aspects as summarized in section 6.3.

This paper is dedicated to the memory of CEUR-WS board member Ralf Klamma † January 2023.

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8.3 Declaration of use of Generative AI

The authors used AI-based coding assistants during the engineering and preparation of this work. Use of such assistants is governed by an AGENTS.md policy file maintained in this paper's repository and, equivalently, in the source repositories of the software artefacts ceur-spt [13] and pyCEURmake [12]. The policy mandates:

– a plan-and-confirm protocol: the assistant must state its analysis, plan, and scope, and receive explicit human confirmation before any read, write, or execution;
– a strict scope for the manuscript itself: AI assistance is limited to spelling/grammar corrections and formatting support (tables, figures, build automation). AI was not used to generate, rewrite or contribute original scientific content, arguments, or prose;

All scientific claims, interpretations and conclusions are the sole responsibility of the authors.

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